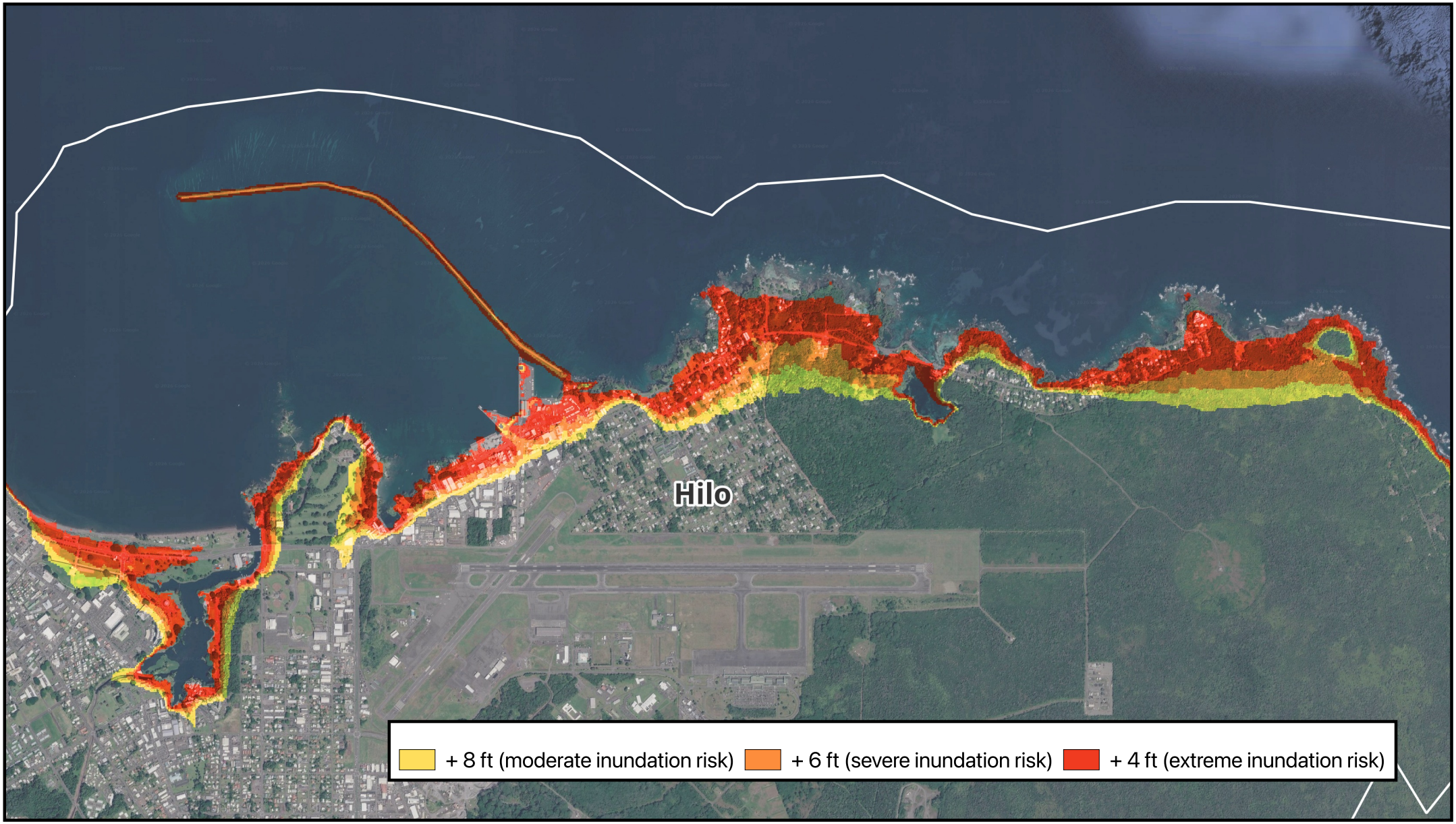
Sea Level Rise and Inundation Risk in Hilo, Hawaii in 2100

Cartography by Sophia Santos



0 1 2 3 4 km

RF Scale: 1 to 35,000

Data Sources: NOAA National Centers for Environmental Information (NCEI) DEMs, U.S. Census Bureau TIGER/Line

Sophia Santos

GEOG 7: Assignment 7 Write-Up

Based on my research, I chose to portray the Sea Level Rise scenario of 4, 6, and 8 feet in a tourist area of Hilo, Hawaii. According to the NASA Interagency Sea Level Rise Scenario Tool, this scenario is relevant to analyze for the years approaching 2100. The graph on the website, titled "Sea Level Rise for Different Sea Level Scenarios," depicts 4, 6, and 8 feet elevation areas as intermediate, intermediate high, and high risk, respectively, for the year 2100 (NASA). This timeline is particularly relevant as it marks the beginning of the next century. This is a time that this generation's children will live in.

From the NASA Interagency Sea Level Rise Scenario Tool, I learned that many factors contribute to sea level change. Graphs on the website reveal that the current largest contributors to sea level rise by 2030 are Sterodynamic Sea Level Vertical and Land Motion (NASA). By 2100, Antarctica and Greenland will be the biggest contributors to the changes we see in the map (NASA).

In observation of my QGIS map, we see that a majority of the peninsula, which forms the bay by jutting into the coastal waters of Hilo, is at high risk of inundation by 2100. This is evident as the thin strip of land is completely shaded with red and orange, indicating extreme and severe inundation risk. Additionally, a large portion of the coastal perimeter visualized in the map is at moderate, severe, and extreme risk of inundation. Hilo is one of the larger tourist areas in the Big Island of Hawaii as it is surrounded by the many diverse biomes showcasing volcanoes, rainforests, waterfalls, and mountainous coffee farms. The coast is mostly populated with hotels and areas geared towards tourists' stays, thus putting one of Hawaii's main economic drivers at risk. The declining tourism may also help Hawaii protect their natural environments and safeguard residents who inhabit other areas of the island.